

Table of Content

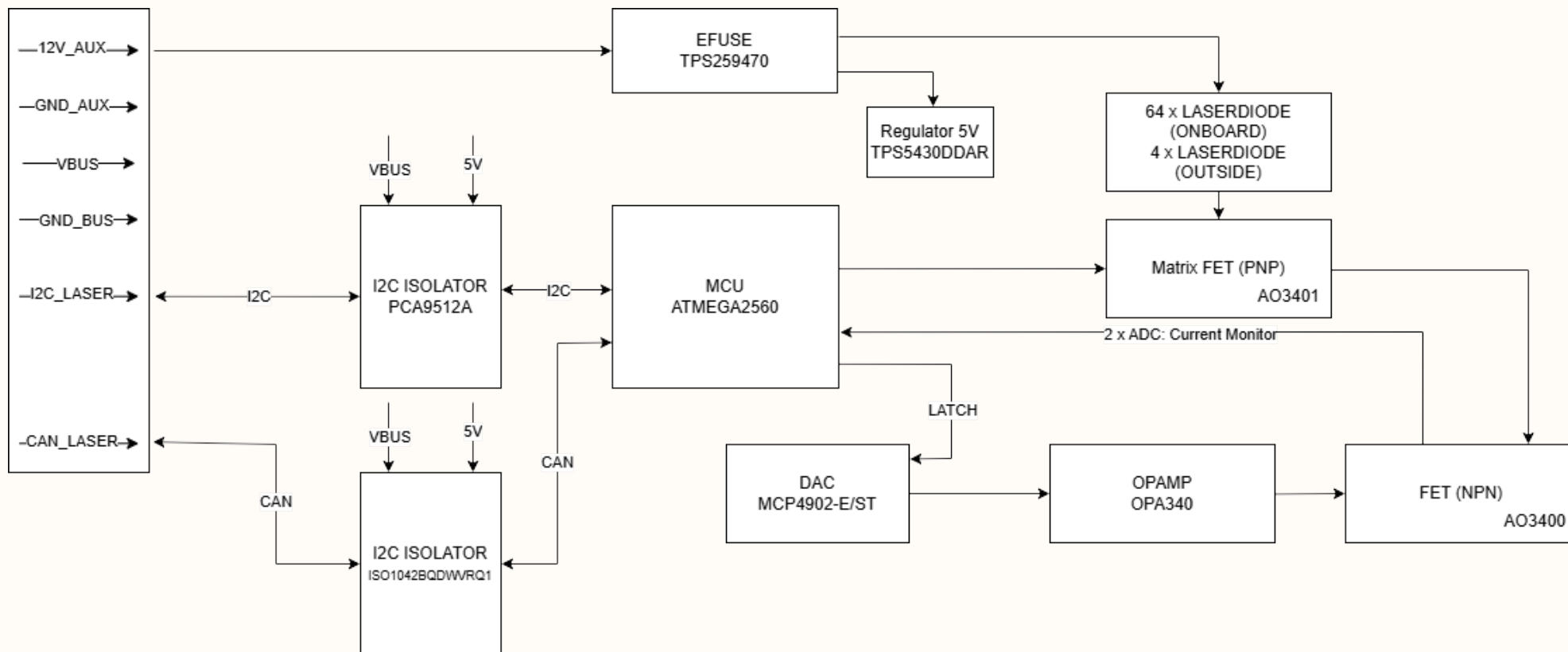
PAGE	COMPONENT/ FUNCTION
001	INDEX
002	BLOCK_DIAGRAM
003	EFUSE_AND_REGULATOR
004	MCU
005	COMMUNICATION
006	CURRENT_SOURCE
007	SWITCHING
008	SWITCHING (CONT)

NANORACK LASER

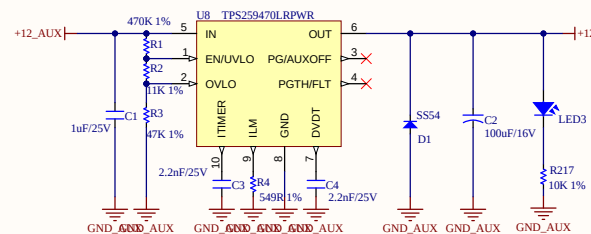
Schematic V1.0.0



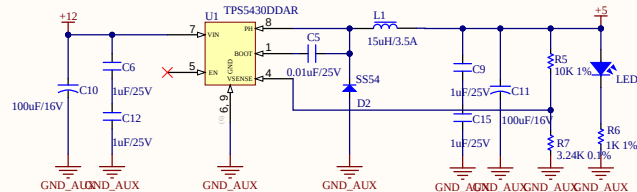
Project: BEE-PC1	Sub-system: NANORACK_LASER.PrjPcb	
Design by: Hung Nguyen D.	Sheet title: INDEX.SchDoc	Sheet 1 of 8
Checked by: Bao Bui Q.	Document No: 1	Size: A3
Approved by: Vu Pham	Revision: V1.0.0	Date: 10/24/2024



Project: BEE-PC1	Sub-system: NANORACK_LASER.PrjPcb	
Design by: Hung Nguyen D.	Sheet title: BLOCK_DIAGRAM.SchDoc	Sheet 2 of 8
Checked by: Bao Bui Q.	Document No: 2	Size: A3
Approved by: Vu Pham	Revision: V1.0.0	Date: 10/28/2024



EFUSE



REGULATOR

$$SR\left(\frac{V}{ms}\right) = \frac{V_{IN}(V)}{t_R(ms)}$$

$$C_{avdt}(pF) = \frac{2000}{SR\left(\frac{V}{ms}\right)}$$

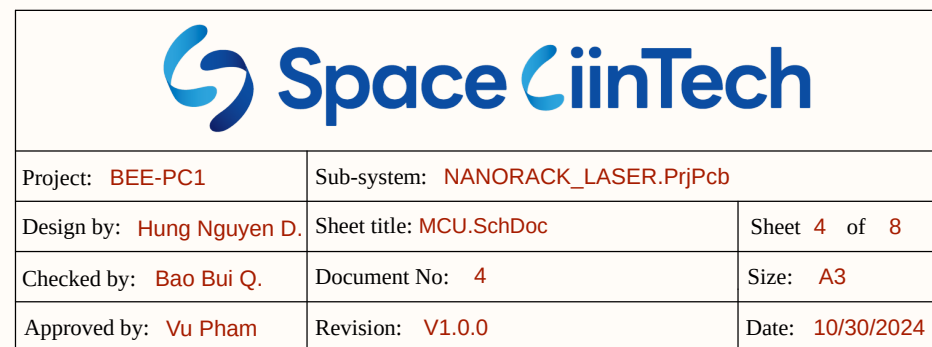
$$I_{INRUSH}(mA) = SR(V/ms) \times C_{OUT}(uF)$$

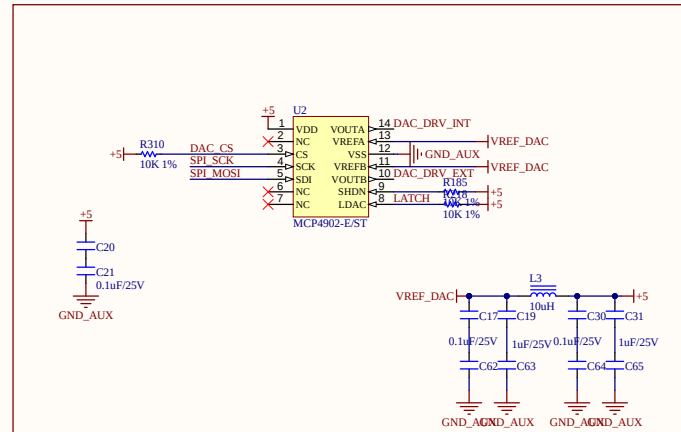
$$R_{LIM}(\Omega) = \frac{3334}{I_{LIM}(A)}$$

$$R_2 = \frac{R_1 \times 1,221}{V_{OUT} - 1,221}$$



Project: BEE-PC1	Sub-system: NANORACK_LASER.PrjPcb	Sheet 3 of 8
Design by: Hung Nguyen D.	Sheet title: EFUSE_AND_REGULATOR.SchDoc	
Checked by: Bao Bui Q.	Document No: 3	Size: A3
Approved by: Vu Pham	Revision: V1.0.0	Date: 10/30/2024

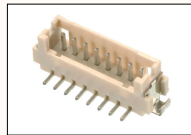




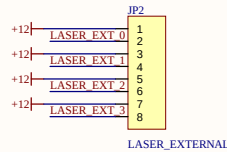
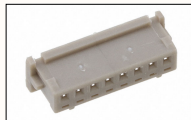
DAC

Harwin Inc. M30-6000846

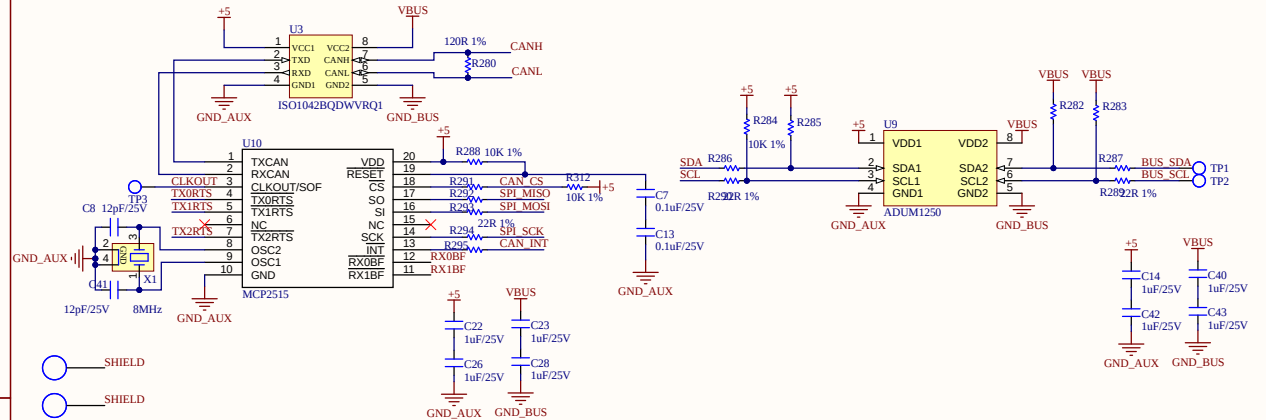
- ELECTRICAL:
CURRENT RATING: 1A/CONTACT
VOLTAGE RATING: 150V/CONTACT
- ENVIRONMENTAL:
TEMPERATURE RANGE: -25°C TO +85°C
- PHYSICAL:
PITCH-MATING: 0.049" (1.25mm)
- MATING:
MAX CURRENT: 1A/CONTACT
Harwin Inc. M30-1100800



MATING

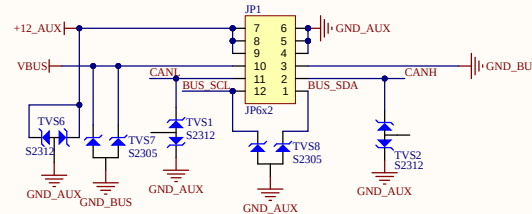


LASER CONNECTOR



CAN

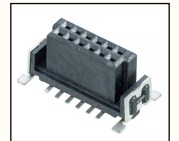
I2C_ISOLATED



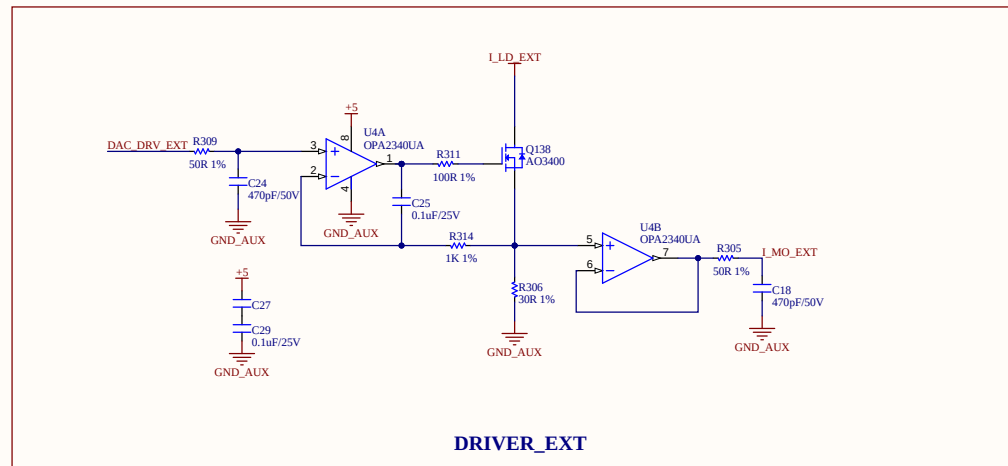
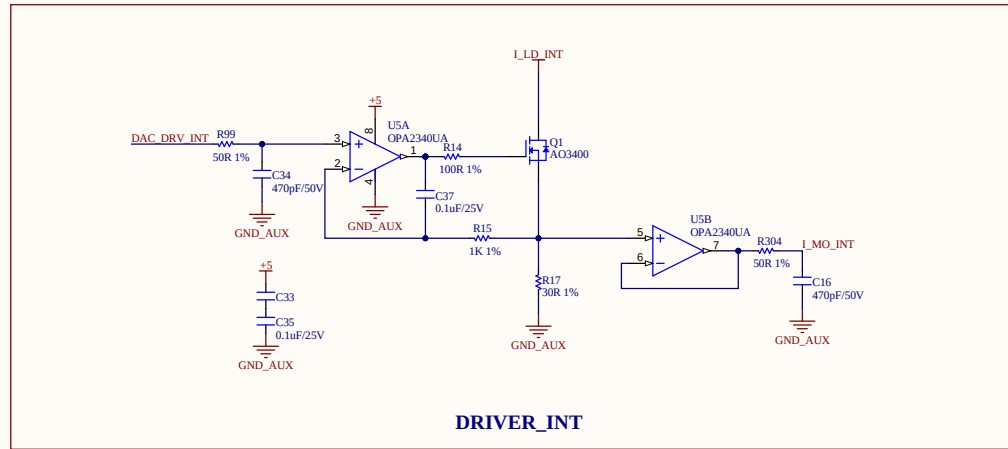
CONNECTOR

Harwin Inc. M55-6001242R

- ELECTRICAL:
CURRENT RATING: 1.2A/CONTACT
VOLTAGE RATING: 100V/CONTACT
- ENVIRONMENTAL:
TEMPERATURE RANGE: -55°C TO +125°C
- PHYSICAL:
PITCH-MATING: 0.050" (1.27mm)
- MATING:
Board to board
MAX CURRENT: 0.7A/CONTACT
Harwin Inc. M55-7001242R



Project: BEE-PC1	Sub-system: NANORACK_LASER.PrjPcb	
Design by: Hung Nguyen D.	Sheet title: COMMUNICATION.SchDoc	Sheet 5 of 8
Checked by: Bao Bui Q.	Document No: 5	Size: A3
Approved by: Vu Pham	Revision: V1.0.0	Date: 10/30/2024



Project: BEE-PC1	Sub-system: NANORACK_LASER.PrjPcb	
Design by: Hung Nguyen D.	Sheet title: CURRENT_SOURCE.SchDoc	Sheet 6 of 8
Checked by: Bao Bui Q.	Document No: 6	Size: A3
Approved by: Vu Pham	Revision: V1.0.0	Date: 10/24/2024

